

Starter Activity

Write a python program to find the area
of the triangle

Formula: $\frac{1}{2}bh$

Python If Statement

Understanding conditions, examples,
and exercises

Introduction to If Statement

The if statement is used for decision making in Python.

It executes a block of code if the given condition is True.

Syntax:

```
if condition:
```

```
    # code block
```

Basic Example

- Example:

```
x = 10
```

```
if x > 5:
```

```
    print("x is greater than 5")
```

Output: x is greater than 5

- Equals: $a == b$
- Not Equals: $a != b$
- Less than: $a < b$
- Less than or equal to: $a <= b$
- Greater than: $a > b$
- Greater than or equal to: $a >= b$

If-Else Statement

The else statement executes a block of code if the condition is False.

Example:

```
x = 3
```

```
if x > 5:
```

```
    print("x is greater than 5")
```

```
else:
```

```
    print("x is not greater than 5")
```

If-Elif-Else Statement

Used when multiple conditions need to be checked.

Example:

```
a = int(input("Enter your first number"))
b = int(input("Enter your second Number"))
if b > a:
    print("b is greater than a")
elif a == b:
    print("a and b are equal")
else:
    print("a is greater")
```

Nested If Statement

An if statement inside another if statement.

Example:

```
x = 15
```

```
if x > 10:
```

```
    print("Above 10")
```

```
    if x > 20:
```

```
        print("Above 20")
```

```
    else:
```

```
        print("Not above 20")
```

Write a Python program that asks the user to enter a number and checks whether the number is **zero**, **positive**, or **negative**.

Logic Operator

a = 200

b = 33

c = 500

if a > b and c > a:

 print("Both conditions are True")

Exercises

1. Write a program to check if a number is even or odd.
2. Write a program to check if a person is eligible to vote (age ≥ 18).
3. Write a program to find the largest of three numbers.
4. Write a program to check if a number is positive, negative, or zero.